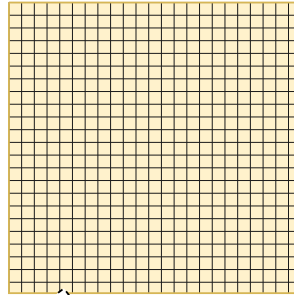


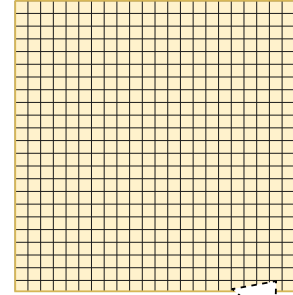
Problem Grid

Finest grid



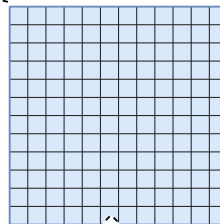
Level Operators
Smooth $Ax^h = b^h$
Residual $r^h = b^h - Ax^h$

Level 0

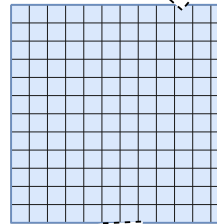


Level Operators
Smooth $Ax^h = b^h$
Error correction $x^h += l^h$

Coarsening Operator
Restrict $b^{2h} = \text{restrict}(r^h)$



Level 1



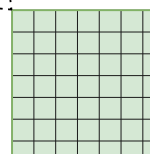
Interpolation Operator
 $l^h = \text{interpolate}(x^{2h})$

Level Operators
Smooth $Ax^{2h} = b^{2h}$
Residual $r^{2h} = b^{2h} - Ax^{2h}$

Level Operators
Smooth $Ax^{2h} = b^{2h}$
Error correction $x^{2h} += l^{2h}$

Coarsening Operator
Restrict $b^{4h} = \text{restrict}(r^{2h})$

Level 2



Coarsest grid

Interpolation Operator
 $l^{2h} = \text{interpolate}(x^{4h})$

Coarsest level
Direct/iterative solver
Solve $Ax^{4h} = b^{4h}$